

Gouki Minegishi

Ph.D. Candidate · The University of Tokyo

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RESEARCH INTERESTS

Reasoning

Interpretability

Safety

EDUCATION

Ph.D. in Computer Science

Apr 2025 – Mar 2028

The University of Tokyo · Advisor: Prof. Yutaka Matsuo

M.S. in Computer Science

Apr 2023 – Mar 2025

The University of Tokyo

B.S. in Engineering

Apr 2019 – Mar 2023

The University of Tokyo

SELECTED PUBLICATIONS

01 Emergent Analogical Reasoning in Transformers

ICML 2026

Gouki Minegishi, Jingyuan Feng, Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

International Conference on Machine Learning (ICML), 2026 **SPOTLIGHT (TOP 2.2%)**

02 Steering at the Source: Style Modulation Heads for Robust Persona Control

ICML 2026

Yoshihiro Izawa, Gouki Minegishi, Koshi Eguchi, Sosuke Hosokawa, and Kenjiro Taura

International Conference on Machine Learning (ICML), 2026

03 Understanding Emergent Misalignment via Feature Superposition Geometry

ACL 2026

Gouki Minegishi, Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

Annual Meeting of the Association for Computational Linguistics (ACL), 2026

04 Mechanism of Task-oriented Information Removal in In-context Learning

ICLR 2026

Hakaze Cho, Haolin Yang, Gouki Minegishi, and Naoya Inoue

International Conference on Learning Representations (ICLR), 2026

05 RL Squeezes, SFT Expands: A Comparative Study of Reasoning LLMs

ICLR 2026

Kohsei Matsutani, Shota Takashiro, Gouki Minegishi, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

International Conference on Learning Representations (ICLR), 2026

06 LIT-RAGBench: Benchmarking Generator Capabilities of Large Language Models in Retrieval-Augmented Generation

LREC-COLING 2026

Koki Itai, Shunichi Hasegawa, Yuta Yamamoto, Gouki Minegishi, and Masaki Otsuki

Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING), 2026

07 In-Context Neurofeedback: Can LLMs Control Their Internal Representations through Privileged Access?

EMNLP Findings 2026

Koshiro Aoki, Ryota Takatsuki, Gouki Minegishi, Yusuke Haruki, and Daisuke Kawahara

Findings of the Association for Computational Linguistics: EMNLP, 2026

Trustworthy AI for Good (AI4GOOD) Workshop @ ICML, 2026 **SPOTLIGHT**

08 Zipping the Thought: When and How Compressed Reasoning Data Works in LLM Post-Training

EMNLP Findings 2026

Kohsei Matsutani, Gouki Minegishi, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo

Findings of the Association for Computational Linguistics: EMNLP, 2026

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Topology of Reasoning: Understanding Large Reasoning Models through Reasoning Graph Properties
Gouki Minegishi, Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo
Neural Information Processing Systems (NeurIPS), 2025

NeurIPS 2025

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Beyond Induction Heads: In-Context Meta Learning Induces Multi-Phase Circuit Emergence
Gouki Minegishi, Hiroki Furuta, Shohei Taniguchi, Yusuke Iwasawa, and Yutaka Matsuo
International Conference on Machine Learning (ICML), 2025
ICLR Workshop on Building Trust in Language Models and Applications, 2025 ORAL

ICML 2025

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Rethinking Evaluation of Sparse Autoencoders through the Representation of Polysemous Words
Gouki Minegishi, Hiroki Furuta, Yusuke Iwasawa, and Yutaka Matsuo
International Conference on Learning Representations (ICLR), 2025

ICLR 2025

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Interpreting Multi-Attribute Confounding through Numerical Attributes in Large Language Models
Hirohane Takagi*, Gouki Minegishi*, Shota Kizawa, Issey Sukeda, Hitomi Yanaka (* Equal Contribution)
International Joint Conference on NLP / Asia-Pacific Chapter of ACL (IJCNLP-AACL), 2025

IJCNLP-AACL 2025

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Bridging Lottery Ticket and Grokking: Understanding Grokking from Inner Structure of Networks
Gouki Minegishi, Yusuke Iwasawa, and Yutaka Matsuo
Transactions on Machine Learning Research (TMLR)

TMLR

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Towards Empirical Interpretation of Internal Circuits and Properties in Grokked Transformers on Modular Polynomials
Hiroki Furuta, Gouki Minegishi, Yusuke Iwasawa, and Yutaka Matsuo
Transactions on Machine Learning Research (TMLR)
ICLR Workshop on Bridging the Gap Between Practice and Theory in Deep Learning, 2024 ORAL

TMLR

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ADOPT: Modified Adam Can Converge with Any β_2 with the Optimal Rate
Shohei Taniguchi, Keno Harada, Gouki Minegishi, et al.
Neural Information Processing Systems (NeurIPS), 2024

NeurIPS 2024

■ RESEARCH & WORK EXPERIENCE

Student Researcher	May 2026 – Oct 2026
Google DeepMind, Japan	
Research Scientist	Jun 2025 – Present
Third Intelligence, Inc.	
Research on developing new forms of intelligence beyond conventional AI, aiming to bridge human-like reasoning, perception, and action.	
Chief AI Engineer	Nov 2024 – May 2025
Matsuo Institute, Inc.	
Developed automated driving software, including 3D object detection with deep learning and integration of the World Model into automated driving.	
Research Intern	Aug 2024 – Oct 2024
Preferred Networks, Inc.	
Research on in-context learning in foundation models, with knowledge conflicts.	

■ ACADEMIC HONORS

Best Paper Award, JSAI 2026	2026
Spotlight Presentation (top 2.2%), ICML 2026	2026
Spotlight Presentation, ICML 2026 AI4GOOD Workshop	2026
Young Researcher Encouragement Award, NLP	2025

- Outstanding Presentation Award, JSAI 2024 2024
- Oral Presentation, ICLR 2025 Workshop (Building Trust in LMs) 2025
- Oral Presentation, ICLR 2024 Workshop (Bridging Practice and Theory) 2024
- Selected for "BOOST NAIS" Scholarship Program for fostering advanced AI talents

■ ACADEMIC SERVICE

Reviewer:

- ICML — 2026 (Silver Reviewer), 2025
- NeurIPS — 2026, 2025
- ICLR — 2026
- AAAI — 2026
- TMLR

Organizer: Mechanistic Interpretability Organized Session at JSAI 2025